MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY ALLAHABAD

MINI PROJECT REPORT

GAN-Based Medical Image Generation for Skin Disease Classification

Mini Project (CSN17351)

Submitted by

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Abstract

This mini project investigates whether class-conditioned synthetic skin-lesion images generated by a Conditional Generative Adversarial Network (CGAN) can support a downstream skin-disease image-classification task. A CGAN was trained to generate 64 × 64 RGB images conditioned on one of nine disease classes. The project generated 20 synthetic images per class (180 images in total) and compared two CNN classifier training conditions: real images only, and real images augmented with the generated images.

The available split contained 697 training images and 181 validation images. The real-only classifier achieved 29.28% final validation accuracy and a macro F1-score of 0.2223. The real-plus-synthetic classifier achieved 28.73% final validation accuracy and a macro F1-score of 0.2315. Synthetic augmentation therefore did not improve final validation accuracy in this experiment, although macro F1 increased slightly. Generated samples were visually blurry, and label agreement measured using the baseline classifier was 21.11% (38 of 180). These results are preliminary and the system is not intended for clinical use.

Keywords

Conditional GAN; medical image synthesis; skin lesion classification; data augmentation; convolutional neural network; synthetic data evaluation.

1. Introduction

Skin conditions can exhibit substantial visual variation, while labelled medical-image datasets may be limited in size or unevenly distributed across diagnostic categories. Generative models offer a way to produce additional image samples. However, generated images are not automatically realistic, diverse, or correctly labelled. Their usefulness must be assessed through sample inspection and downstream-task evaluation.

This project explores a class-conditional image-generation workflow and evaluates whether adding generated samples to the classifier's training data changes validation performance. It is an experimental study, not a clinical diagnostic system.

1.1 Problem Statement

To investigate whether synthetic skin-disease images generated by a CGAN can augment a real-image training set and improve a CNN classifier's performance across nine skin-disease classes.

1.2 Objectives

• Prepare a nine-class image dataset.

• Train a CGAN conditioned on class labels.

• Generate 20 synthetic images per class (180 total).

• Train CNN classifiers on real-only and real-plus-synthetic data.

• Compare validation accuracy and macro-averaged metrics.

• Document limitations and avoid unsupported clinical claims.

2. Dataset

The project used the Kaggle dataset “Skin Disease Classification Image Dataset” by riyaelizashaju, using the Split_smol subset. The reported working split contained 697 training images and 181 validation images. No independent test folder was available.

Class-index order: 0: Actinic keratosis; 1: Atopic Dermatitis; 2: Benign keratosis; 3: Dermatofibroma; 4: Melanocytic nevus; 5: Melanoma; 6: Squamous cell carcinoma; 7: Tinea Ringworm Candidiasis; 8: Vascular lesion.

Dataset source: Kaggle dataset page.

3. Methodology

3.1 Overall workflow

The workflow consists of dataset preparation, CGAN training, class-conditional image generation, sample/label-consistency checks, CNN classifier training under two data conditions, and validation-based comparison.

3.2 Conditional GAN

The generator receives a latent noise vector of dimension 128 together with a one-hot class vector of dimension 9 and produces a 64 × 64 RGB image. The discriminator is trained to distinguish real from generated images while using the class condition. The adversarial objective encourages generated samples to resemble the training distribution while respecting the requested class condition.

The project generated 20 samples for each of nine classes, for 180 synthetic images. The generated images were observed to be blurry and to lack fine visual detail.

3.3 CNN classifier

The downstream classifier is a compact CNN with three Conv2D blocks, global average pooling, a dense layer, dropout, and a nine-class softmax output. It contains 102,089 parameters. Both conditions used the same classifier design and were trained for 10 epochs.

3.4 Experimental conditions

Condition	Training data	Count
Real-only	Real images	697
Real + synthetic	697 real + 180 synthetic	877

The augmented run initially encountered an input-pipeline exhaustion warning. It was corrected using a repeated dataset and steps_per_epoch=28.

4. Evaluation Metrics

Final validation accuracy is the fraction of validation examples classified correctly at the end of the run. Macro F1 is the unweighted mean of class-wise F1 scores. Macro precision and recall similarly average class-level precision and recall. No confidence intervals or repeated-run estimates are available.

5. Experimental Results

Metric	Real-only	Real + synthetic
Final validation accuracy	29.28%	28.73%
Macro F1-score	0.2223	0.2315
Best validation accuracy	32.60% (epoch 6)	31.49% (epoch 7)
Final validation loss	1.6967	1.7739

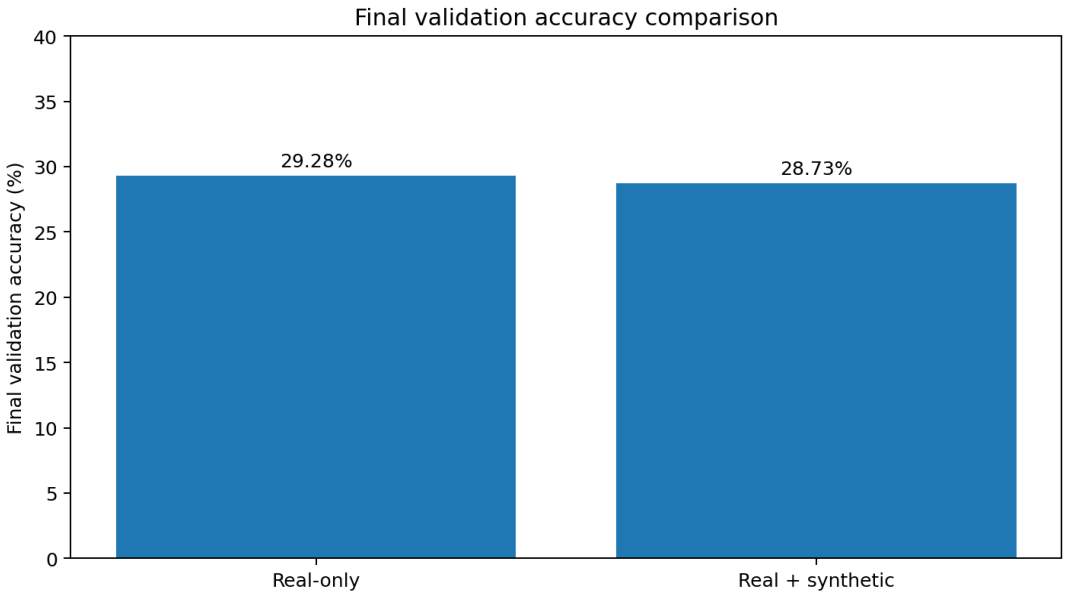


Figure 1. Final validation accuracy comparison.

5.1 Synthetic sample label agreement

A baseline classifier agreed with the intended labels for 38 of 180 generated images (21.11%). This is only a weak proxy, not an independent ground-truth assessment. Because the baseline classifier performed poorly, disagreement is not definitive proof that a generated image has the wrong class.

5.2 Interpretation

Final validation accuracy decreased by 0.55 percentage points after adding synthetic images, while macro F1 increased by 0.0092. The results do not support a claim that synthetic augmentation improved overall accuracy. The saved classifier files represent final-epoch weights, not best-validation-epoch checkpoints.

6. Limitations

• Small dataset: 697 training and 181 validation images.

• No independent test set.

• Generated 64 × 64 images were visibly blurry.

• Only one run per condition; no statistical significance analysis.

• Label agreement relied on a weak baseline classifier, not expert annotation.

• Saved classifier weights are final-epoch rather than best-validation checkpoints.

• No external clinical validation or dermatologist review.

7. Conclusion

This project implemented a class-conditioned GAN pipeline and evaluated adding synthetic images to a CNN classifier's training set. The real-plus-synthetic condition did not improve final validation accuracy (28.73% versus 29.28% for real-only), although macro F1 increased slightly (0.2315 versus 0.2223). Given the dataset size, blurry samples, weak label-agreement proxy, and lack of an independent test set, the evidence is insufficient to conclude that synthetic images provide a dependable benefit.

Future work should improve image resolution and generator stability, obtain expert review, use a larger and more diverse dataset, preserve an independent test set, save best-validation checkpoints, and repeat experiments across multiple random seeds.

8. Reproducibility and Repository

The repository contains notebooks, trained generator and classifier models, and experiment result files. Intended flow: obtain the dataset from Kaggle, prepare the expected class-folder structure, run the CGAN notebook, generate synthetic samples, and run the evaluation pipeline. Exact package versions, runtime settings, and paths should be verified against the notebooks before claiming full environment reproducibility.

Project GitHub repository.

9. References

• Goodfellow, I., et al. (2014). Generative Adversarial Nets. Advances in Neural Information Processing Systems.

• Mirza, M., & Osindero, S. (2014). Conditional Generative Adversarial Nets. arXiv:1411.1784.

• riyaelizashaju. Skin Disease Classification Image Dataset. Kaggle.

Declaration and Disclaimer

This work is an academic mini-project. The models and outputs have not been clinically validated and are not intended to diagnose, screen, or guide treatment for any person. Performance values are limited to the stated experimental split and setup.